

	Lubrication Instructions			Rev: A	05-05-25 hm
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				Sheets: 7	Sheet: 2
7316.4352.0 Narayangonj Slag		Slag Grinding Plant with LOESCHE-Mill LM 46.2 + 2 S		Substitute for: Revision: Name: Date:	
LOESCHE-Mill: Rocker Arm "M"; cpl. / Rocker Arm "S"; cpl.					
			Number of Groups: 2 each		
Components	Rocker Arm "M" Fixed Bearing	Rocker Arm "M" Floating Bearing	Rocker Arm "S" Bearing		
Number of Components	1	1	1		
Filling Quantity	~ 8 kg filled	~ 6 kg filled	filled (Grease Cartridge)		
First Change	-	-	-		
Further Changes	14'000 h Check Bearings	14'000 h Check Bearings	14'000 h Check Bearings		
Refilling Quantity	see "Remarks"	see "Remarks"	-		
Check Quantity / Refill	see "Remarks"	see "Remarks"	-		
Specifications of Lubricant:	Special Grease	Special Grease	-		
ISO 3448	-	-	-		
DIN 51502 / DIN 51825	KPF 2 G-20	KPF 2 G-20	-		
Viscosity at 40° C [cSt]	-	-	-		
Penetration	265 – 295	265 – 295	-		
Type of Thickener	Li	Li	-		
Lubricant (recommended)	Dow Corning Molykote Longterm 2 Plus	Dow Corning Molykote Longterm 2 Plus	-		
Remarks	Greased by LM- Lubrication Line	Greased by LM- Lubrication Line	Greased for Lifetime of Bearing		
Appertaining Documents (Drawing-no. or Manual)	Rocker Arm (M); cpl. 044230-00-0	Rocker Arm (M); cpl. 044230-00-0	Rocker Arm (S); cpl. 025458-00-0		
LOESCHE-Mill: LM-Lubrication Line: cpl.					
			Number of Groups: 1		
Components	Grease Pump FZ A-B				
Number of Components	1				
Filling Quantity	8 ltr.				
First Change	-				
Further Changes	-				
Refilling Quantity	see Line-Measure				
Check Quantity / Refill	every day				
Specifications of Lubricant:	Special Grease				
ISO 3448	-				
DIN 51502 / DIN 51825	KPF 2 G-20				
Viscosity at 40° C [cSt]	-				
Penetration	265 – 295				
Type of Thickener	Li				
Lubricant (recommended)	Dow Corning Molykote Longterm 2 Plus				
Remarks	Lubrication Cycle per Mill Operation hour: 5 min. ON 55 min. OFF				
Appertaining Documents (Drawing-no. or Manual)	LM-Lubrication Line; cpl. 025810-00-0 and Scheme: Lubrication Line 902274-00-3				

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7316.4352.0 Narayangonj Slag	Slag Grinding Plant with LOESCHE-Mill LM 46.2 + 2 S		Substitute for: Revision: Name: Date:	
LOESCHE-Mill: Grinding Roller "M"; cpl. / Grinding Roller "S"; cpl.				
Number of Groups: 2 each				
Components	Grinding Roller "M" Roller Bearing	Grinding Roller "S" Roller Bearing		
Number of Components	1	1		
Filling Quantity	~ 160 ltr.	~ 235 ltr.		
First Change	300 – 600 h	300 – 600 h		
Further Changes	Check Oil-Quality every 3'500 h	14'000 h		
Refilling Quantity	see "Remarks"	see Oil-Dipstick		
Check Quantity / Refill	see "Remarks"	every month		
Specifications of Lubricant:	Mineral Oil	Synthetic Oil		
ISO 3448	VG 320	VG 460		
DIN 51502 / DIN 51825	CLP 320	PGLP 460		
Viscosity at 40° C [cSt]	320	460		
Penetration	-	-		
Type of Thickener	-	-		
Lubricant (recommended)	Mobil Mobilgear 632	Mobil Glygoyle HE 460		
Remarks	Lubricated by Hydraulic Cabinet HSMS	Sump-Lubrication		
Appertaining Documents (Drawing-no. or Manual)	Roller (M); cpl. 021874-00-0	Roller (S); cpl. 043288-00-0		
LOESCHE-Mill: INTEGRAL-Hydraulic Cabinet HSMS 850				
Number of Groups: 1				
Components	Hydraulic Cabinet Oil-Tank			
Number of Components	1			
Filling Quantity	850 ltr.			
First Change	300 – 600 h			
Further Changes	Check Oil-Quality every 3'500 h			
Refilling Quantity	see Line-Measure			
Check Quantity / Refill	every day			
Specifications of Lubricant:	Mineral Oil			
ISO 3448	VG 320			
DIN 51502 / DIN 51825	CLP 320			
Viscosity at 40° C [cSt]	320			
Penetration	-			
Type of Thickener	-			
Lubricant (recommended)	Mobil Mobilgear 632			
Remarks	HSMS lubricates Roller Bearings of Rollers "M"			
Appertaining Documents (Drawing-no. or Manual)	INTEGRAL-Supplier Documentation			

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7316.4352.0 Narayangonj Slag		Slag Grinding Plant with LOESCHE-Mill LM 46.2 + 2 S		Substitute for: Revision: Name: Date:	
LOESCHE-Mill: Spring Assembly "M"; cpl.				Number of Groups: 2	
Components	- Hydraulic Cylinders - Hydr. -Accumulators - Pipings	Hydraulic Cylinders Rod Eye Bearings: (Cylindrical Roller Bearing)	Hydraulic Cylinders Rod Eye Bearings: (Tilting Bearings)	Hydraulic Cylinders: Bottom Eye Bearing:	
Number of Components	2 Cylinders each 4 Piston Accus each 8 Bladder Accus each	1 each (Grease Nipple)	2 each (Grease Nipples)	1 each (Grease Nipple)	
Filling Quantity	~ 400 ltr.	~ 0.5 kg	~ 1.5 kg	filled	
First Change	-	During inspection of bearings	During inspection of bearings	During inspection of bearings	
Further Changes	Check Oil-Quality every 3'500 h	14'000 h Check Bearings	14'000 h Check Bearings	14'000 h Check Bearings	
Refilling Quantity	see "Remarks"	50 g	200 g	50 g	
Check Quantity / Refill	see "Remarks"	every 14 days of operation	every 3 months of operation	every 3 months of operation	
Specifications of Lubricant:	Hydraulic Oil	Special Grease	Special Grease	Special Grease	
ISO 3448	VG 68	-	-	-	
DIN 51502 / DIN 51825	HLP 68	KPF 2 K-10	KPF 2 K-10	KPF 2 K-10	
Viscosity at 40° C [cSt]	68	-	-	-	
Penetration	-	265 – 295	265 – 295	265 – 295	
Type of Thickener	-	Li / Ca	Li / Ca	Li / Ca	
Lubricant (recommended)	Mobil DTE 26	Use only SKF-Grease LGEV 2	Use only SKF-Grease LGEV 2	Use only SKF-Grease LGEV 2	
Remarks	Lubricated by Hydraulic Cabinet HSLM	Greased by Hand	Greased by Hand	Greased by Hand	
Appertaining Documents (Drawing-no. or Manual)	Spring Assembly (M); cpl. 024540-00-0	Spring Assembly (M); cpl. 024540-00-0	Spring Assembly (M); cpl. 024540-00-0	Spring Assembly (M); cpl. 024540-00-0	
LOESCHE-Mill: INTEGRAL-Hydraulic Cabinet HSLM 1000				Number of Groups: 1	
Components	Hydraulic Cabinet Oil-Tank				
Number of Components	1				
Filling Quantity	1000 ltr.				
First Change	-				
Further Changes	Check Oil-Quality every 3'500 h				
Refilling Quantity	see Line-Measure				
Check Quantity / Refill	every day				
Specifications of Lubricant:	Hydraulic Oil				
ISO 3448	VG 68				
DIN 51502 / DIN 51825	HLP 68				
Viscosity at 40° C [cSt]	68				
Penetration	-				
Type of Thickener	-				
Lubricant (recommended)	Mobil DTE 26				
Remarks	HSLM lubricates Hydraulic System "M"				
Appertaining Documents (Drawing-no. or Manual)	INTEGRAL-Supplier Documentation				

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7316.4352.0 Narayangonj Slag	Slag Grinding Plant with LOESCHE-Mill LM 46.2 + 2 S		Substitute for: Revision: Name: Date:	
LOESCHE-Mill: Spring Assembly "S"; cpl.				
			Number of Groups: 2	
Components	- Hydraulic Cylinder - Pipings	Hydraulic Cylinder: Rod Eye Bearing	Hydraulic Cylinder: Bottom Eye Bearing	
Number of Components	1 Cylinder each	1 each (Grease Nipple)	1 each (Grease Nipple)	
Filling Quantity	~ 20 ltr.	filled	filled	
First Change	-	During inspection of bearings	During inspection of bearings	
Further Changes	Check Oil-Quality every 3'500 h	14'000 h Check Bearings	14'000 h Check Bearings	
Refilling Quantity	see "Remarks"	50 g	50 g	
Check Quantity / Refill	see "Remarks"	every 3 months of operation	every 3 months of operation	
Specifications of Lubricant:	Hydraulic Oil	Special Grease	Special Grease	
ISO 3448	VG 68	-	-	
DIN 51502 / DIN 51825	HLP 68	KPF 2 K-10	KPF 2 K-10	
Viscosity at 40° C [cSt]	68	-	-	
Penetration	-	265 – 295	265 – 295	
Type of Thickener	-	Li / Ca	Li / Ca	
Lubricant (recommended)	Mobil DTE 26	Use only SKF-Grease LGEV 2	Use only SKF-Grease LGEV 2	
Remarks	Lubricated by Hydraulic Cabinet HSSW	Greased by Hand	Greased by Hand	
Appertaining Documents (Drawing-no. or Manual)	Spring Assembly (S); cpl. 025462-00-0	Spring Assembly (S); cpl. 025462-00-0	Spring Assembly (S); cpl. 025462-00-0	
LOESCHE-Mill: INTEGRAL-Hydraulic Cabinet HSSW 100				
			Number of Groups: 1	
Components	Hydraulic Cabinet Oil-Tank			
Number of Components	1			
Filling Quantity	100 ltr.			
First Change	-			
Further Changes	Check Oil-Quality every 3'500 h			
Refilling Quantity	see Line-Measure			
Check Quantity / Refill	every day			
Specifications of Lubricant:	Hydraulic Oil			
ISO 3448	VG 68			
DIN 51502 / DIN 51825	HLP 68			
Viscosity at 40° C [cSt]	68			
Penetration	-			
Type of Thickener	-			
Lubricant (recommended)	Mobil DTE 26			
Remarks	HSSW lubricates Hydraulic System "S"			
Appertaining Documents (Drawing-no. or Manual)	INTEGRAL-Supplier Documentation			

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7316.4352.0 Narayangonj Slag		Slag Grinding Plant with LOESCHE-Mill LM 46.2 + 2 S		Substitute for: Revision: Name: Date:	
LOESCHE-Mill: RENK-Mill Gearbox LGP 1120-46 CS					
				Number of Groups: 1	
Components	Oil-Tank	Positioning Drive Nord SK 9032-90L			
Number of Components	1	1			
Filling Quantity	~ 5'200 ltr.	1.7 ltr.			
First Change	~ 600 h (Oil Analysis)	after 10'000 h or after 2 years			
Further Changes	every 4'000 h or every 6 month (Oil Analysis) exchange every 20'000 h at least	every 10'000 h or every 2 years			
Refilling Quantity	see Line-Measure	see Line-Measure			
Check Quantity / Refill	every 100 h	every month			
Specifications of Lubricant:	Mineral Oil	Mineral Oil			
ISO 3448	VG 320	VG 220			
DIN 51502 / DIN 51825	CLP 320	CLP 220			
Viscosity at 40° C [cSt]	320	220			
Penetration	-	-			
Type of Thickener	-	-			
Lubricant (recommended)	Mobil Mobilgear XMP 320	Mobil Mobilgear 630			
Remarks	-	-			
Appertaining Documents (Drawing-no. or Manual)	RENK-Supplier Documentation	RENK-Supplier Documentation			
LOESCHE-Mill: Swinging Device with RUHFUS HSAV 125				Number of Groups: 1	
Components	Pump Trolley HSAV 125	Swinging Cylinder			
Number of Components	1	1			
Filling Quantity	114 ltr.	~ 255 ltr.			
First Change	check Oil-Quality every 2 years	check Oil-Quality every 2 years			
Further Changes	check Oil-Quality every 2 years	check Oil-Quality every 2 years			
Refilling Quantity	see Line-Measure	by HSAV			
Check Quantity / Refill	before use	before use			
Specifications of Lubricant:	Hydraulic Oil	Hydraulic Oil			
ISO 3448	VG 68	VG 68			
DIN 51502 / DIN 51825	HLP 68	HLP 68			
Viscosity at 40° C [cSt]	68	68			
Penetration	-	-			
Type of Thickener	-	-			
Lubricant (recommended)	Mobil DTE 26	Mobil DTE 26			
Remarks	-	Cylinder shall be stored filled and in vertical Position			
Appertaining Documents (Drawing-no. or Manual)	RUHFUS-Supplier Documentation	Swinging Device; cpl. 021877-00-0			

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7316.4352.0 Narayangonj Slag		Slag Grinding Plant with LOESCHE-Mill LM 46.2 + 2 S		Substitute for: Revision: Name: Date:		
LOESCHE-Classifer: LSKS 66 CS with FLENDER-Gearbox						Number of Groups: 1
Components	Bearing Cartridge: - Upper Bearing - Lower Bearing	Classifier Lubrication Line: Grease Pump FZ A-B	Classifier Drive: FLENDER-Gearbox B2 SV 07			
Number of Components	1	1	1			
Filling Quantity	~ 2 kg	8 ltr.	~ 35 ltr.			
First Change	-	-	400 h			
Further Changes	-	-	5'000 h or every 18 month			
Refilling Quantity	2 cm³ / h	see Line-Measure	see Line-Measure			
Check Quantity / Refill	see "Remarks"	every day	every month			
Specifications of Lubricant:	Special Grease	Special Grease	Synthetic Oil			
ISO 3448	-	-	VG 320			
DIN 51502 / DIN 51825	KPF 2 G-20	KPF 2 G-20	PGLP 320			
Viscosity at 40° C [cSt]	-	-	320			
Penetration	265 – 295	265 – 295	-			
Type of Thickener	Li	Li	-			
Lubricant (recommended)	Dow Corning Molykote Longterm 2 Plus	Dow Corning Molykote Longterm 2 Plus	FUCHS Gearmaster PGP 320			
Remarks	Greased by Classifier- Lubrication Line	Lubrication Cycle per Mill Operation hour: 5 min. ON 55 min. OFF	Check also name- plate of gearbox for oil-viscosity and quantity			
Appertaining Documents (Drawing-no. or Manual)	Bearing Cartridge; cpl. 049334-00-1	Classifier Lubrication; cpl 047960-00-1 and Scheme: Lubrication 902228-00-4	FLENDER-Supplier Documentation			
WAGNER: Magnetic Machine 0446-30/70						Number of Groups: 1
Components	Drum Bearings	Gearboxmotor SEW: SA47DT80K				
Number of Components	2	1				
Filling Quantity	workshop filled	~ 0.35 ltr. workshop filled				
First Change	-	after 10'000 h or after 2 years				
Further Changes	-	every 10'000 h or every 2 years				
Refilling Quantity	2.5 g each	see Line-Measure				
Check Quantity / Refill	1'500 h	every 3'000 h				
Specifications of Lubricant:	Grease	Mineral Oil				
ISO 3448	-	VG 220				
DIN 51502 / DIN 51825	KP 2 N-30	CLP 220				
Viscosity at 40° C [cSt]	-	220				
Penetration	279	-				
Type of Thickener	Li	-				
Lubricant (recommended)	Mobil Mobilux EP 2	Mobil Mobilgear 630				
Remarks	-	Check also name-plate of gearbox for oil-viscosity and quantity				
Appertaining Documents	WAGNER-Supplier Manual	WAGNER-Supplier Manual				